Klaudio Fusha & Ethan Shanbaum

Final Project Brainstorming

Proposal

1. Visualizing how people from different states tie certain colors to temperature
    a. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0091854
    b. Objective:
        i. We would ask users what state they are from, then they will answer many stimuli where they are given a specific temperature in degrees Fahrenheit and will pick a color from a color wheel that they associate with the temperature. From this we can determine if people from different areas of the country have different color associations for temperature.
    c. Data:
        i. No data from the internet would be involved
            1. We would have to collect the data
    d. Visualization design:
        i. We can create gradients along a temperature scale to show responses from individuals, with lowest temperatures on the left and highest on the right. For aggregating the results, we can show both the average colors as well as the most common.
    e. Anticipated challenges:
        i. Creating gradients that don't just end up looking like rainbows
        ii. We'll need enough people from different states to participate in our study to actually create grounded conclusions
2. Visualizing horse racing data
    a. Objective:
        i. We would analyze the best-performing horses for each of the races and see their sires, trainers, and jockeys in hopes of finding the best conditions to produce the optimal racing horse.
    b. Data:
        i. https://horseracingdatasets.com/racing/
        ii. https://docs.google.com/spreadsheets/d/1ahggu4Z09OxpIEA7cDUxdiTuG5vsKhpzig3J8hrjwu8/edit?gid=0#gid=0
    c. Visualization design:
        i. A cycleable visualization using a drill-down technique, where we would first observe trainer-level trends, then horse-level, then sire-level, etc.
    d. Anticipated challenges:
        i. Dealing with missing data

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Core Process Book Sections

Overview and Motivation: Provide an overview of the project goals and the motivation for it. Consider that this will be read by people who did not see your project proposal.

- The main goal and/or motivation of our project is to determine what conditions (jockey, trainer, sire, etc.) would produce the best racing horse. We think that having a detailed visualization that can properly communicate the optimal conditions would open new doors in either sports betting or recreational horse racing.

Related Work: Anything that inspired you, such as a paper, a web site, visualizations we discussed in class, etc.

- Horse racing data analytics: What data matters most for betting: https://www.twinspires.com/edge/racing/education/horse-racing-data-analytics-what-data-matters-most-for-betting/

Questions: What questions are you trying to answer? How did these questions evolve over the course of the project? What new questions did you consider in the course of your analysis?

- Initial questions:
  - Which horses have the best chances of victory?
  - What factors contributed to those horses being better over others?
    - Did the following contribute?
      - Trainer?
      - Jockey?
      - Genetic lineage (sire)?
- Question evolution:
  - What do the average horse speeds look like based on different factors?
    - Trainer?
    - Jockey?
    - Sire?
- New questions sparked by analysis:
  - Do the contributing factors for horse performance have any correlation to one another? (i.e. Do horses that have good trainers also tend to have good jockeys?)

Data: Source, scraping method, cleanup, etc.

- Source(s):
  - https://horseracingdatasets.com/racing/
  - https://docs.google.com/spreadsheets/d/1ahggu4Z09OxpIEA7cDUxdiTuG5vsKhpzig3J8hrjwu8/edit?gid=0#gid=0
- Data preparation:
  - Average horse speeds:

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        - We initially created a CSV of the original data containing a new column for each horse that represented the horse race ID (did not end up using)
        - Created a CSV containing all the starters and their corresponding information (average speed, trainers, sires, and jockeys)
        - Created another CSV containing only the top 200 starters by average speed and their corresponding information (the actual average speed, trainers, sires, and jockeys)
    - Jockey:
        - Created a CSV containing all the jockeys, the average speeds of the horses they rode, and the actual horses they rode
        - Created another CSV containing the same information, but sorted by average speed
    - Sire:
        - Just like with the average horse speeds data preparation, we initially created a CSV of the original data containing a new column for each horse that represented the horse race ID (did not end up using)
        - Created a CSV containing all the sires, the average speeds of their offspring, and their actual offspring (all sorted by average speed)
    - Trainer:
        - Created a CSV containing all the trainers, the average speeds of the horses they trained, and the actual horses they trained
        - Created another CSV containing the same information, but sorted by average speed

Exploratory Data Analysis: What visualizations did you use to initially look at your data? What insights did you gain? How did these insights inform your design?

- We used some basic Matplotlib charts for explanatory data analysis
    - How many horses are dirt vs. turf
    - How common each distance is
    - **Both these charts helped us determine if there were any outliers in the data and how feasible it would be to split up our analysis into different categories (based on surface and distance)
- Insights:
    - Approximately 2/3rds of races were held on dirt tracks while 1/3 were held on turf tracks. We decided against splitting up results by surface in order to reduce the number of charts needed to explore the data, as well as to ensure there was cohesion among all the horse data for analysis.

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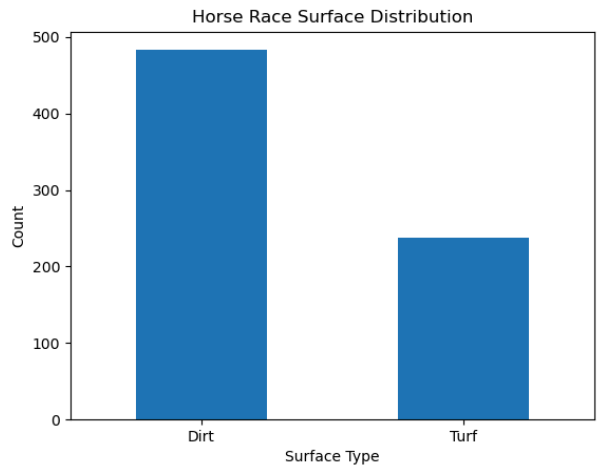
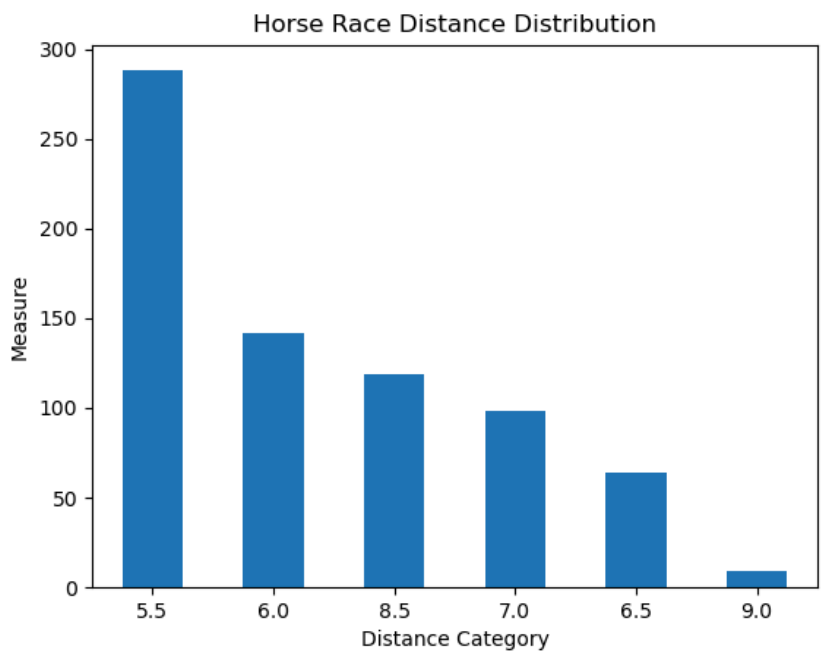


- The most common race length was 5.5 furlongs, with descending frequencies of 6, 8.5, 7, and 6.5 furlongs. 9 furlongs was the rarest race length. Due to the number of different categories, we decided against bucketing our results by race length in order to have a more complete picture of the cumulative data.



- Informing design:
  - Although these preliminary visualizations did not inform our main visualization, they did help us decide on focusing more on uncovering potential trends between definitive horse attributes/characteristics opposed to ones influenced by environment.

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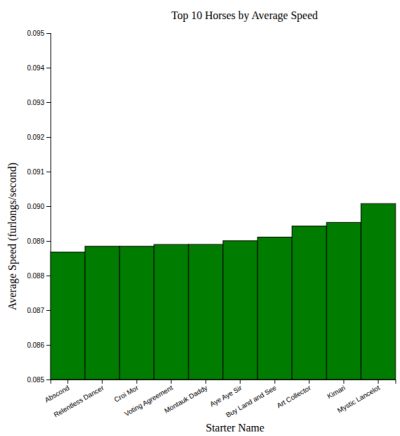
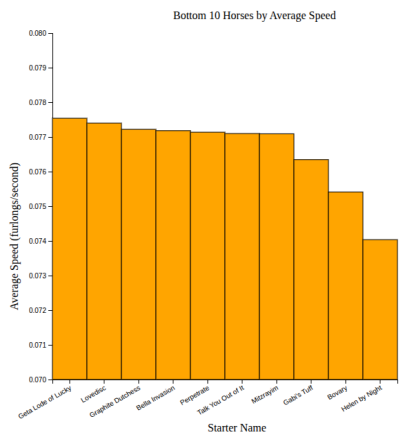
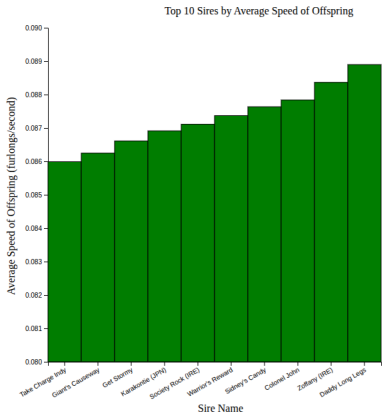
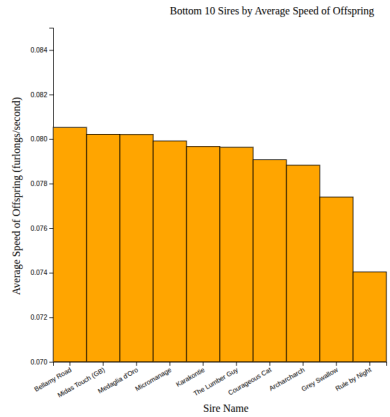
Design Evolution: What are the different visualizations you considered? Justify the design decisions you made using the perceptual and design principles you learned in the course. Did you deviate from your proposal?

- We decided on bar charts from the very start for our preliminary visualizations, and did not think of any other type of chart. This was primarily due to our main goal of just visualizing the data that we prepared and seeing what initial trends we could observe.
- We thought bar charts were a good first choice because they appear to be cleaner in design and provide a nice balance between accuracy and ease of implementation (based on our A3 experiment).
- For the preliminary visualizations, we did deviate from our proposal to some extent. Instead of a pure cycleable visualization using a drill-down technique, we decide to display all the charts at once and couple each one with an information pane (which appears upon hovering). However, we planned to include a feature to connect the different visualizations (average speeds, sires, trainers, jockeys).









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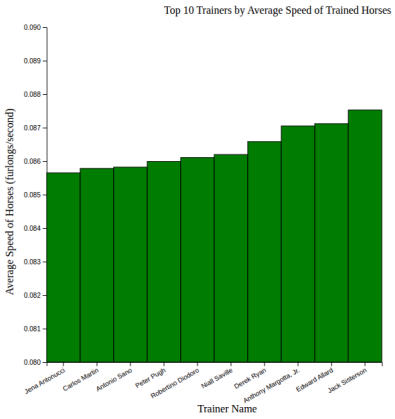

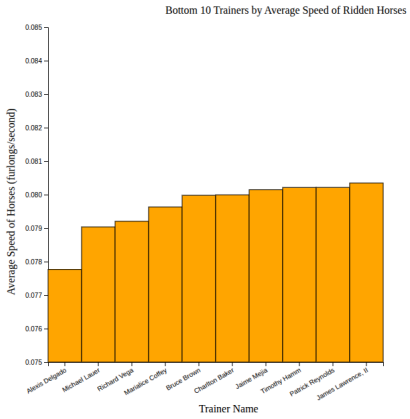

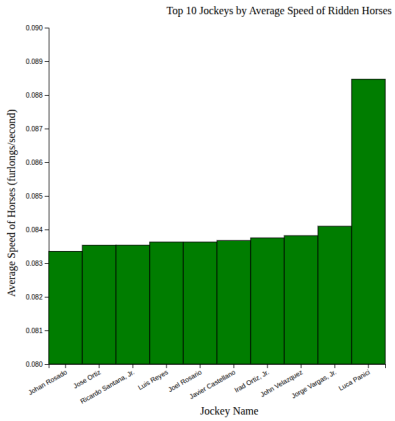

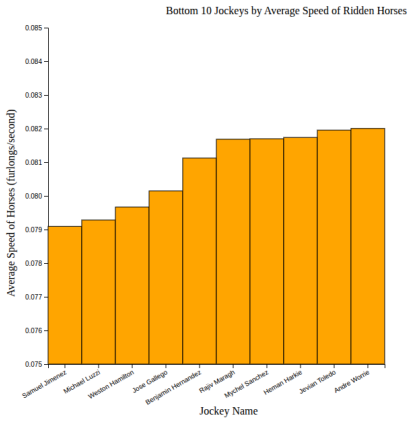

- For the bar graphs displaying the top 10 and bottom 10 speeds for the various variables, we allowed the user to hover over a bar to get more information pertaining to a specific piece of data. This allowed for better understanding of each data and can allow the user to draw connections between the different graphs.

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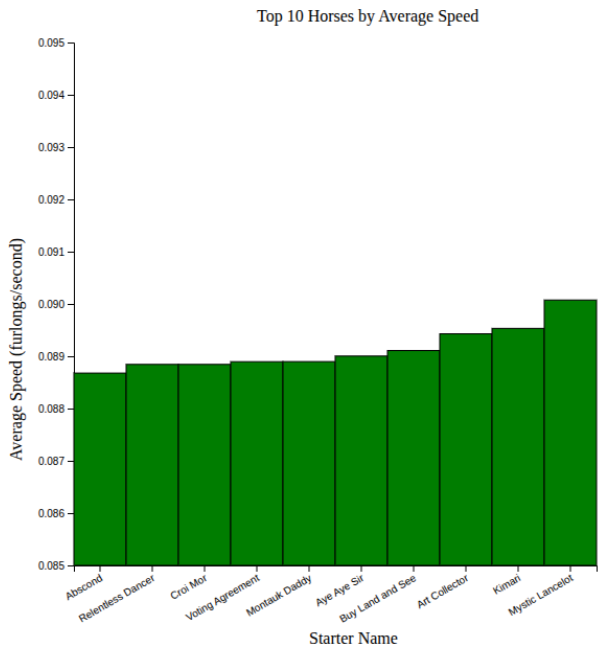

Average Speed (furlongs/second): 0.09007533573534228, Trainer(s): Todd Pletcher, Sire(s): Into Mischief, Jockey(s): ['John Velazquez']

- For the preliminary visualizations (top 10 and bottom 10), we could not find any meaningful patterns. Regarding the average horse speeds chart(s), we needed to expand our ranges for the top X chart and exclude the bottom X chart entirely. For the rest of the charts, since there is a relatively “small” number of examples, we decided to display everything (no shortened ranges were needed)

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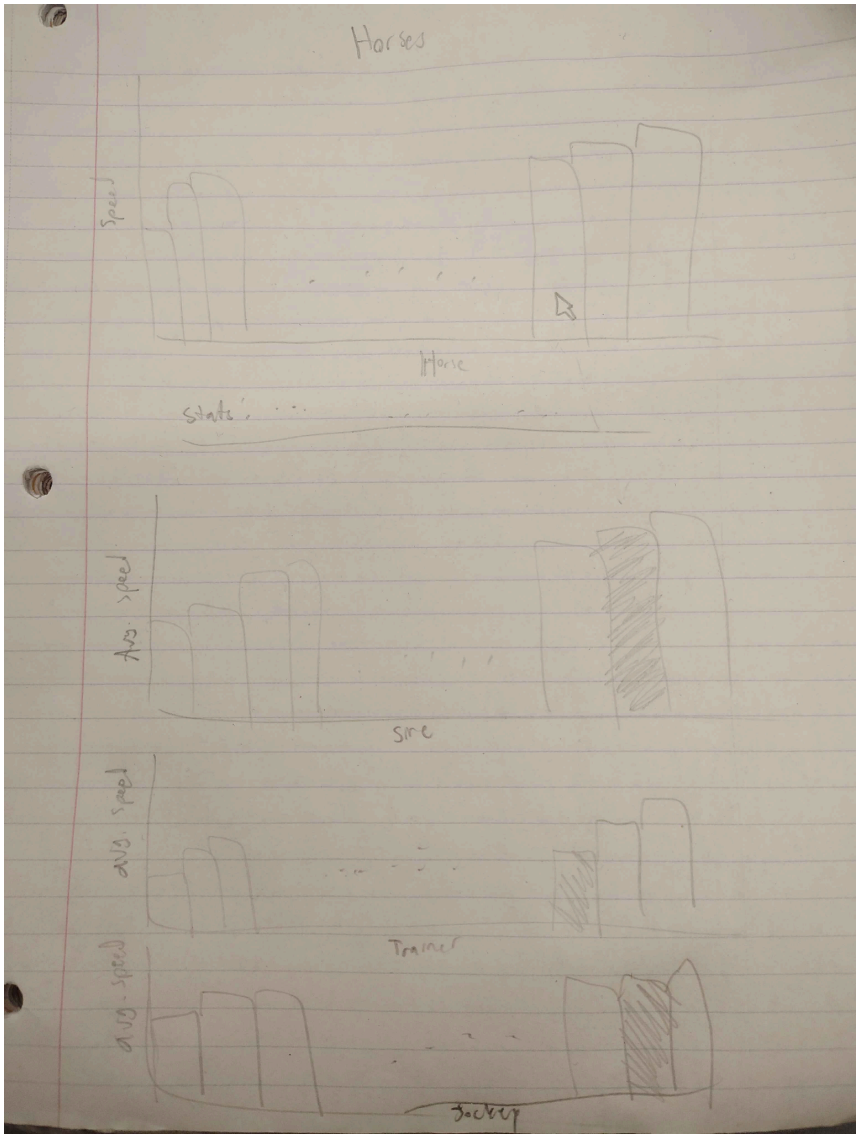

- The above drawing shows the idea we have for updating the visuals. Instead of showing the top 10 and bottom 10, we thought it would be wise to show all of the sires, trainers, and jockeys. After showing our drawing of the updated visualization to the professor and gaining approval, we implemented it in code. We experimented with how many horses to show and decided that the top 200 horses were a good balance between showing plenty of data as well as being comfortable to interact with. With this sketch, we redesigned our visualization:

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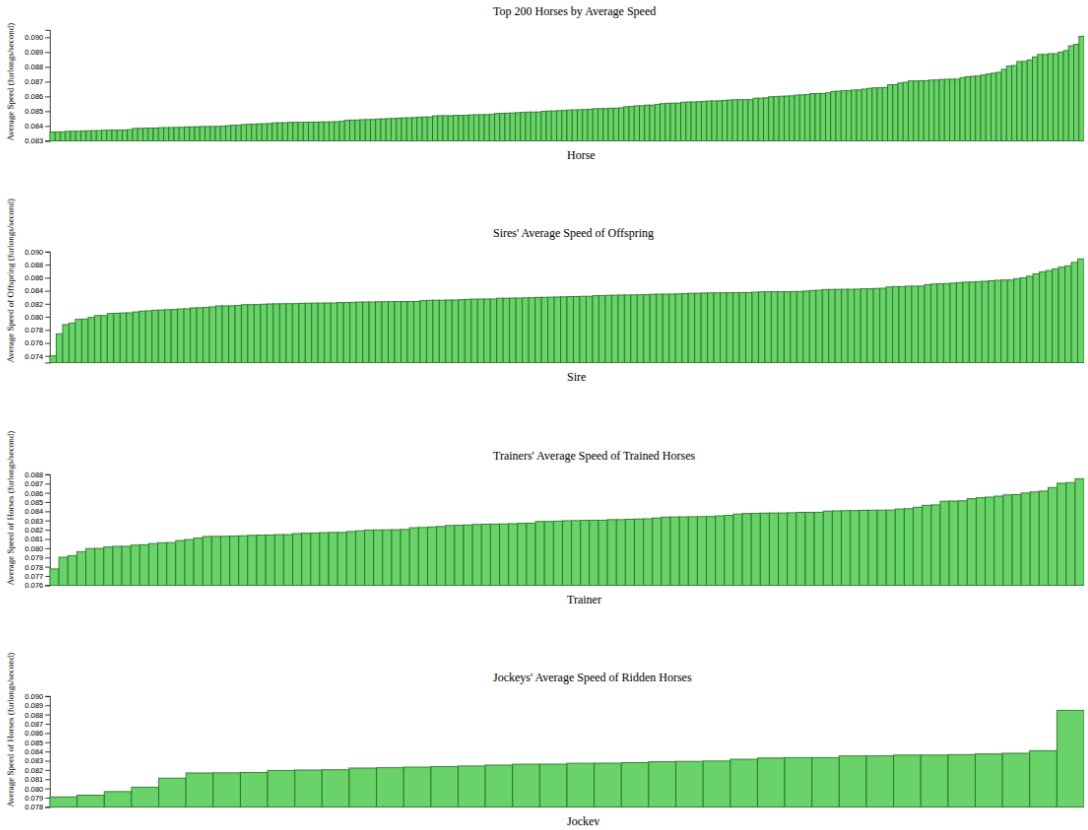

- We went with light-green bars for all four charts and scaled everything to fit more easily on a user's screen, since we plan on adding a feature to allow users to draw connections between the charts via clicking.

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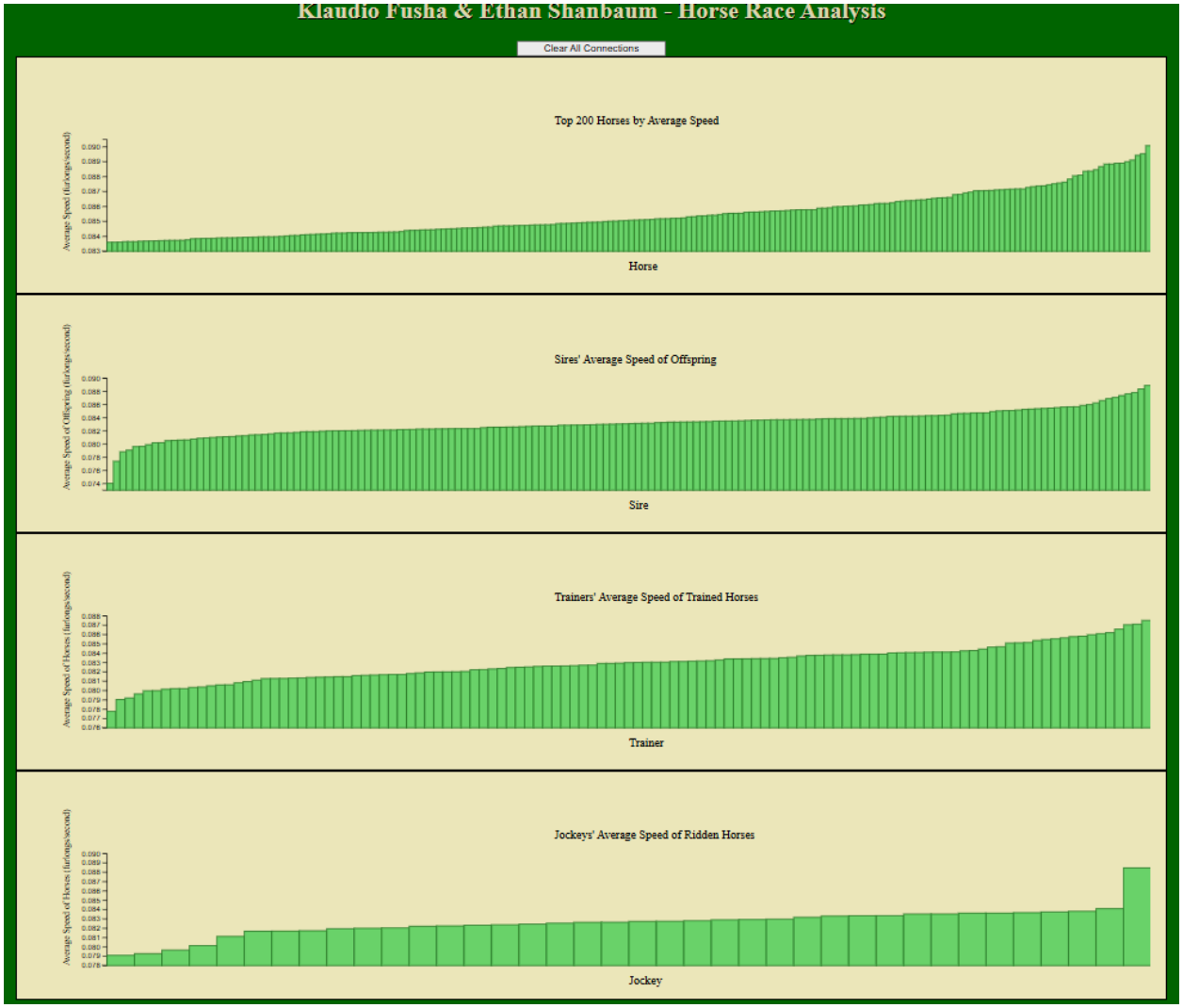

- We then added stylistic elements to give the app a fancier vibe. Also added cross-graph hovering and a horse information pane. Keep in mind, at this point, the hover info pane functionality was still present (hovering over a given bar and relevant information about it being shown right under the graph).

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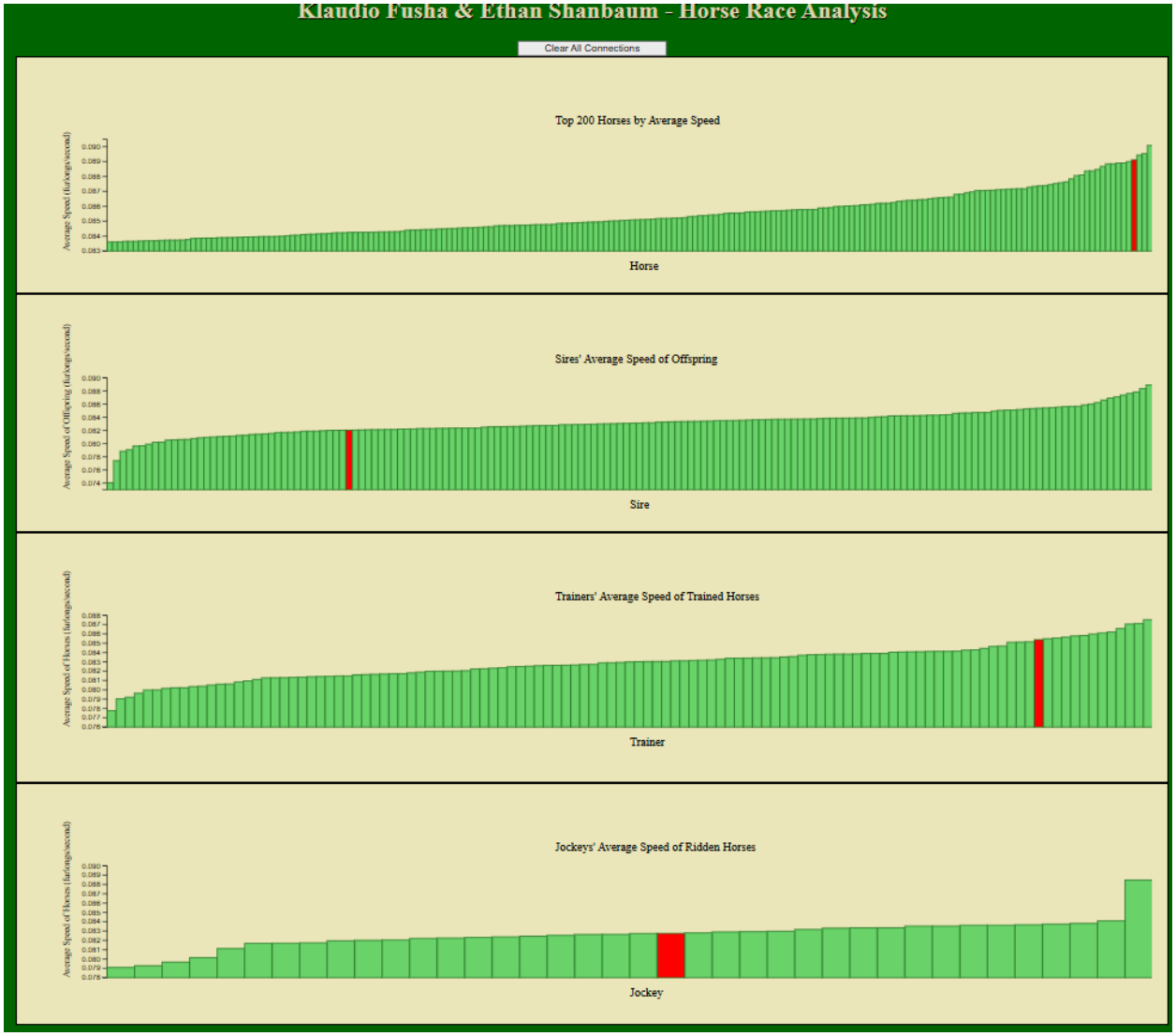

Horse Name	Buy Land and See
Average Speed (furlongs/second)	0.08910923171640582
Trainer(s)	Steve Klesaris
Sire(s)	Cairo Prince
Jockey(s)	['Jose Lezcano']

- After styling was complete, we implemented hovering and clicking functionality to interact with the data. When you click on a bar in the top 200 horses data, the horses information would be written to a table below all of the graphs. This would allow the user to keep track of the horses information while browsing the rest of the data.

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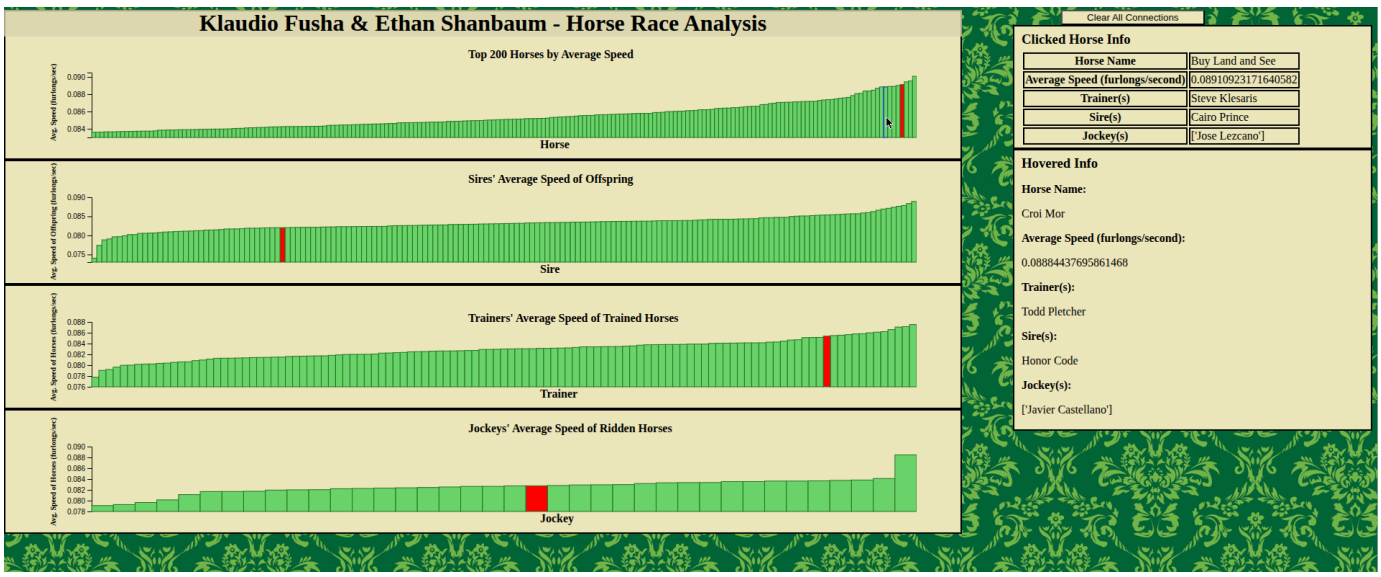
- Additionally, clicking on any of the 4 charts would show the existing data connections. By clicking on a horse data point, the trainers, sire, and jockeys that have been affiliated with that horse will be highlighted in red. This can help us analyze what factors make the best racing horses the fastest. By clicking on the other 3 charts, it will highlight the horses associated with that data. For example, clicking on a jockey would highlight all the horses that the jockey has ridden in a race.
- Hovering over a piece of data will add a blue stroke around that data, to provide visual clarity. The information for the hovered data also appears below the graph the user is looking at.



- After reviewing with Professor Harrison, we decided that having all the charts visible on the screen at once would make analysis easier, since you wouldn't have to scroll down to see some connections with horses that may be offscreen. In order to do this, we redesigned the website to have all 4 charts appear on the screen by shrinking their height. Additionally, we moved the hover info pane and horse information pane to the right of the graph to make all the information available on a single screen for easier analysis.
    - This brought up a technical issue relating to screen height. We wanted to make sure the charts would size dynamically to the users screen, but our charts had a hardcoded width and height. For the width, we were able to dynamically size the charts to fit the width of the user's screen. However, for height, we ran into issues with the axis formatting correctly. In the end, we decided to make the charts fit on a 1920x1080 screen, since this is considered a standard screen size.
- We also added a patterned background for a contrast between the solid backgrounds for the graphs and info panes.

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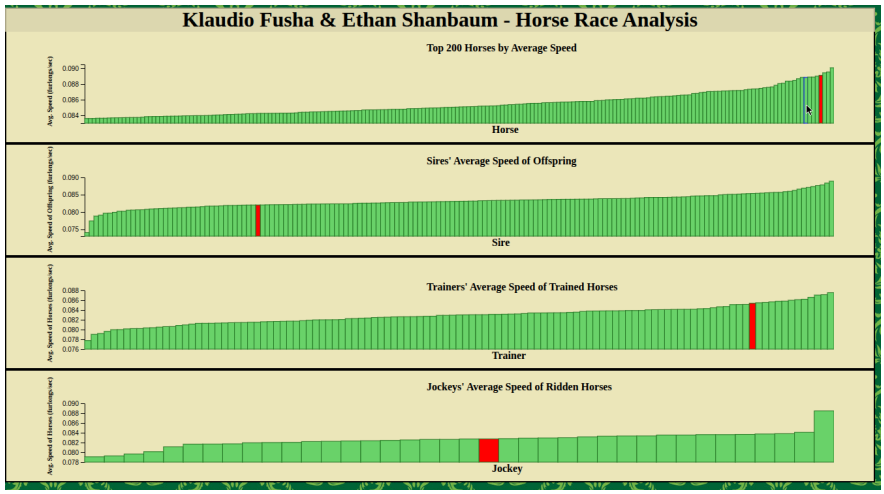
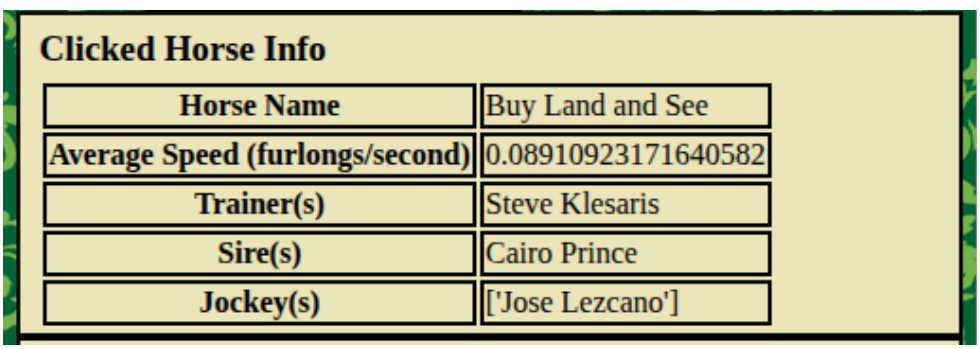
Implementation: Describe the intent and functionality of the interactive visualizations you implemented. Provide clear and well-referenced images showing the key design and interaction elements.

- Our intent with the interactive charts is to allow the user to find connections that exist between the top 200 horses and their trainers, sires, and jockeys. This can allow for analysis as to how these factors can affect the horse's average speed during races.
- The clicking functionality is the main way to display connections, as discussed in the Design Evolution section. The connections drawn are displayed in red to be visually distinct and clear to the user.



- Clicking on a horse will also fill out the horse pane on the right side of the screen. We created a specific information pane just for horse data to ensure the data is not lost when the user hovers off of the information. This could also be used for cross-analysis by being able to compare 2 different horses, the one that is clicked on and the one that is being hovered over.



Clicked Horse Info

Horse Name	Buy Land and See
Average Speed (furlongs/second)	0.08910923171640582
Trainer(s)	Steve Klesaris
Sire(s)	Cairo Prince
Jockey(s)	['Jose Lezcano']

- The hovering functionality allows the user to get a more clear idea of what the data represents. By hovering over the data, the information pane on the right side is filled out with specific info for that data, such as name, average speed, and associated horses.

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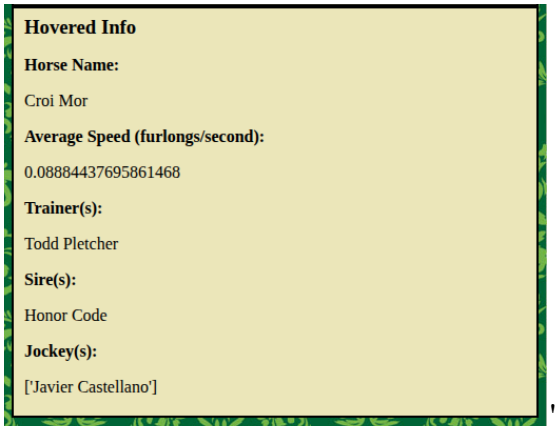

Evaluation: What did you learn about the data by using your visualizations? How did you answer your questions? How well does your visualization work, and how could you further improve it?

- Observations from clicking from right to left on all the charts within the main visualization:
    - Both trainer and sire seem to have an almost linear relationship with the fastest horses (at least for the top trainers and sires)
        - Somewhere between the 50-75% mark of these charts, the trained horses of the trainers and offspring of the sires don't appear in the top 200 fastest horses
    - This shows us that trainer and sire may have more of an impact than jockey when it comes to producing an optimal horse, which aligns with what we expect.
        - This could influence horse bettors in regards to how they choose which horses to bet on. By looking at the average speed for the trainer and sire of a horse, the bettor could determine which horses have a higher chance of performing best.
- Questions answered:
    - Initial questions:
        - Which horses have the best chances of victory?
            - In general, the horses with the best chances of victory include those who have similar genetic traits passed down by the top 100 sires, as well as those trained by the top 100 trainers.
        - What factors contributed to those horses being better over others?
            - Did the following contribute?
                - Trainer?
                - Yes, the better trainers tended to be associated with faster horses.
                - Jockey?
                - From our observations, not that much

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- Genetic lineage (sire)?
    - Yes, the better sires tended to be associated with faster horses.
- Question evolution:
    - What do the average horse speeds look like based on different factors?
        - Trainer?
            - Most trainers tend to have the same performance. However, starting at around the top 16th trainer (right to left), there is a linear incline in performance. Starting from the bottom 18th (left to right), we begin to notice a significant decline in the average speed of the trained horses, with the trainer "Alexis Delgado" being much worse compared to the other trainers.
        - Jockey?
            - One jockey, Luca Panici, is significantly better than the rest of the jockeys. Then there is a fairly linear decline until the last 5, where there is a more significant drop in average speed of ridden horses.
        - Sire?
            - Most sires tend to have the same performance. However, starting at around the top 12th sire (right to left), there is a linear incline in performance. Starting from the bottom 9th (left to right), there is a fairly significant decline in average speed of offspring, with the sire "Rule by Night" being much worse compared to the other sires.
- New questions sparked by analysis:
    - Do the contributing factors for horse performance have any correlation to one another? (i.e. Do horses that have good trainers also tend to have good jockeys?)
        - Horses that had the best speeds tended to have at least one exceptional factor, with the other factors ranging from excellent to below average. For example, the horse "Buy Land and See" had an excellent trainer, but an average jockey and below average sire. On the other hand, the horse "Aye Aye Sir" had 3 excellent factors. It seems like there is no correlation between good trainers and good sires.

- Ultimately, the visualization works pretty well, given that we were able to pick up on this information by interacting with it. One major improvement could include finding a way to display all the horse data in hopes of providing insights for worst-performing horses.